

The 2nd Meeting of the Special Interest Group in Non-Equilibrium Molecular Dynamics

Edward Smith, Matthew Borg and James Ewen

06/09/21

Online

Our SIG in Non-Equilibrium Molecular Dynamics (NEMD)

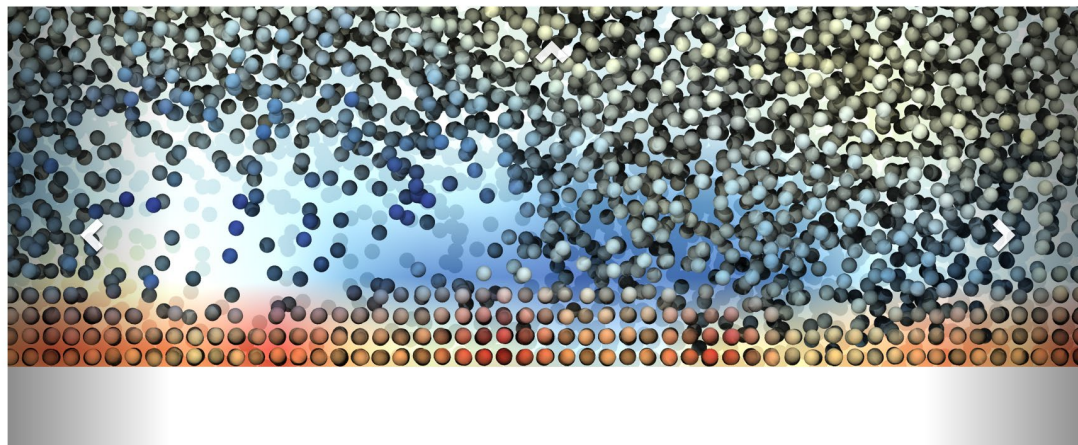
<https://fluids.ac.uk/sig/NonEqmMD> www.fluids.ac.uk/sig/NonEqmMD



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Edward Smith ▾

- Please sign up to the UKFN for future announcements
- The mailing list from this meeting will be used for future meetings and maybe an international NEMD conference



Non-equilibrium Molecular Dynamics (NEW)

Description

Non-equilibrium molecular dynamics (NEMD) is the study of fundamental fluid flow using molecular simulation - 'non-equilibrium' because the system is driven away from thermodynamic equilibrium by wall motion, temperature and pressure gradients or

Tweets

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Nonequilibrium Time Reversibility with Maps and Walks and The Ian Snook Prize

William Graham Hoover, Carol Griswold Hoover, Edward R.
Smith

06/09/21

The Ian Snook Prize

- The Ian Snook Prize (the Prize) honors Australian computational physicist Ian Snook (13 June 1945 – 7 April 2013)
- The Prize is offered by Bill and Carol Hoover (the Organizers)



The Ian Snook Prize

\$1000 SNOOK PRIZES FOR 2021: The Information Dimensions of a Two-Dimensional Baker Map

The fractal information dimension can be computed in three ways:

- (1) mapping points,
- (2) mapping regions (two-dimensional areas here), and
- (3) applying the Kaplan-Yorke conjecture.

For the simplest nonequilibrium Baker N2 Map these three approaches can give different results.

A pedagogical exploration and explanation of this situation is the 2021 Ian Snook Prize Problem.

Background and Link NEMD

What has this got to do with NEMD?

We get Flowfields from Average of Molecules

- Mass

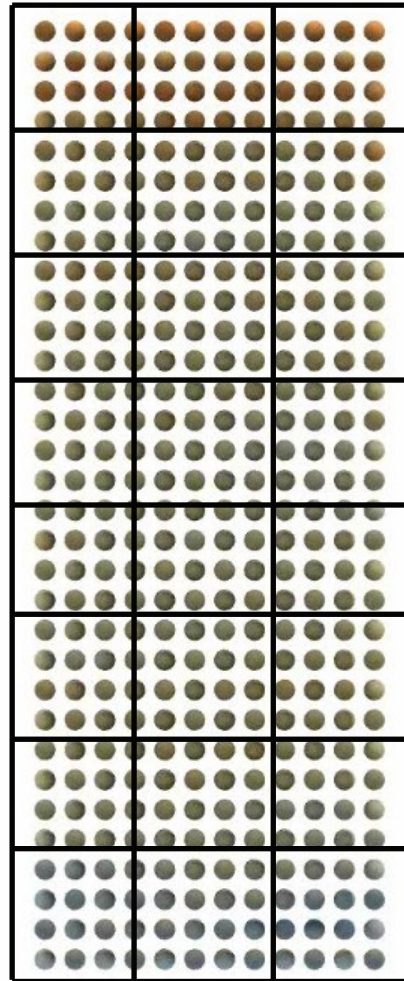
$$\frac{d}{dt} \sum_i m_i = 0$$

- Newton's Law

$$\mathbf{F}_i = m_i \ddot{\mathbf{r}}_i$$

- Energy

$$\frac{d}{dt} \mathcal{H}(q_i, p_i) = 0$$



- Mass

$$\frac{\partial}{\partial t} \int_V \rho dV = - \oint_S \rho \mathbf{u} \cdot d\mathbf{S}$$

- Momentum (Newton's Law/Navier Stokes)

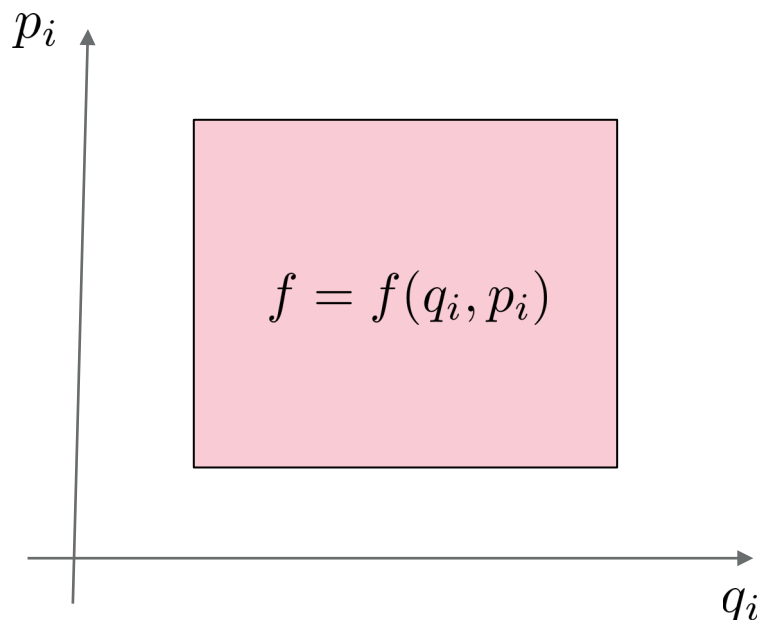
$$\frac{\partial}{\partial t} \int_V \rho \mathbf{u} dV = - \oint_S \rho \mathbf{u} \mathbf{u} \cdot d\mathbf{S} + \mathbf{F}_{\text{surface}}$$

- Energy

$$\begin{aligned} \frac{\partial}{\partial t} \int_V \rho \mathcal{E} dV = & - \oint_S \rho \mathcal{E} \mathbf{u} \cdot d\mathbf{S} \\ & - \oint_S \mathbf{\Pi} \cdot \mathbf{u} \cdot d\mathbf{S} + \mathbf{q} \cdot d\mathbf{S} \end{aligned}$$

Or we use the Phase Space Description

- NEMD can also be understood from statistical mechanics
- Phase space is the N particle system plotted as **Average** positions against momenta of the particles



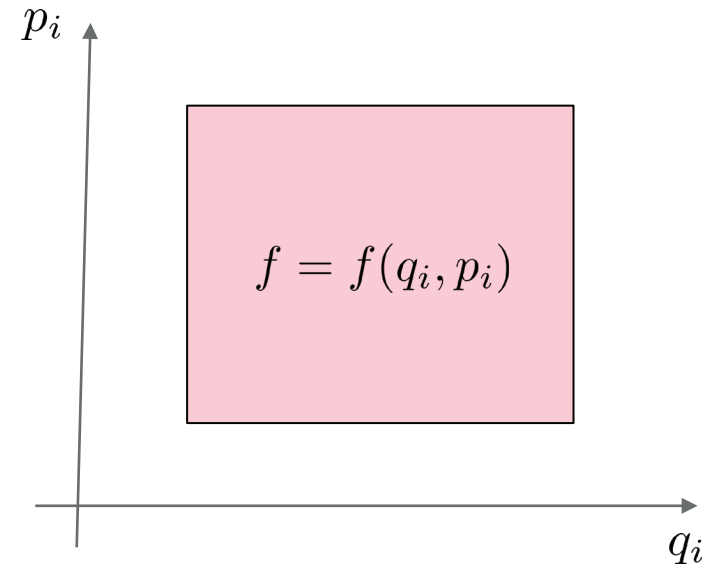
- Any volume in this Phase space is incompressible (like density in the fluid mass continuity equation)
- Liouville's phase-space theorem (time derivative of f)

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_{i=1}^N \left[\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right]$$

Underlying Cause of Many Fluid Quantities

Liouville's Equation

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_{i=1}^N \left[\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right]$$



$$\Pi_{xy} = \underbrace{\sum_{i=1}^N \left\langle m_i v_{xi} v_{yi} dS_{yi}; f \right\rangle}_{\text{Kinetic}} + \underbrace{\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N \left\langle f_{xij} dS_{yij}; f \right\rangle}_{\text{Configurational}} \quad \text{Stress/Pressure}$$

Surface Tension

$$\gamma = \int [\Pi_N - \Pi_T] dr$$

Entropy

$$\frac{P(\dot{S} = A)}{P(\dot{S} = -A)} = e^{At}$$

Viscosity

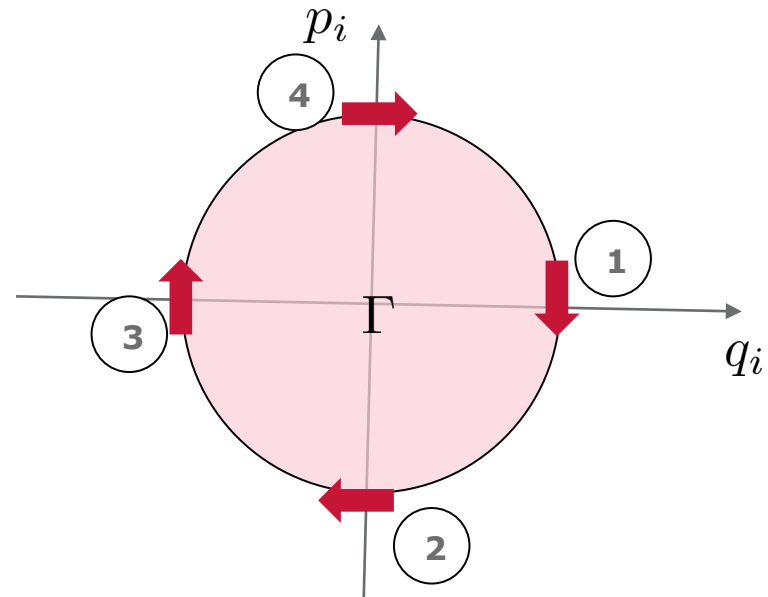
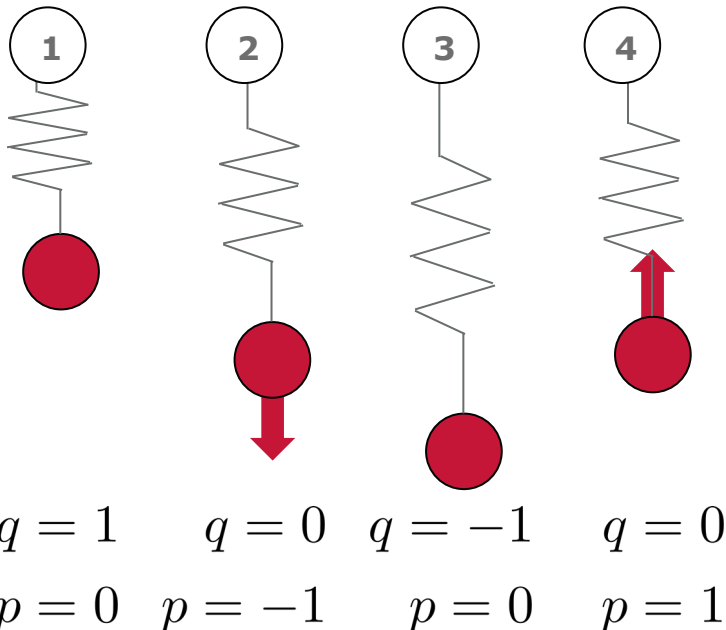
$$\mu = \frac{V}{k_B T} \int_0^t \langle \Pi_{xy}(\tau) \Pi_{xy}(0) \rangle d\tau$$

NEMD Harmonic Oscillator

$$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i} = p_i$$

$$\dot{p}_i = \frac{\partial \mathcal{H}}{\partial q_i} = -q_i$$

$$m_i = k = 1$$



- Phase space area of simple harmonic oscillator is constant

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_{i=1}^N \left[\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right]$$

Nosé-Hoover Oscillator

$$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i} = p_i$$

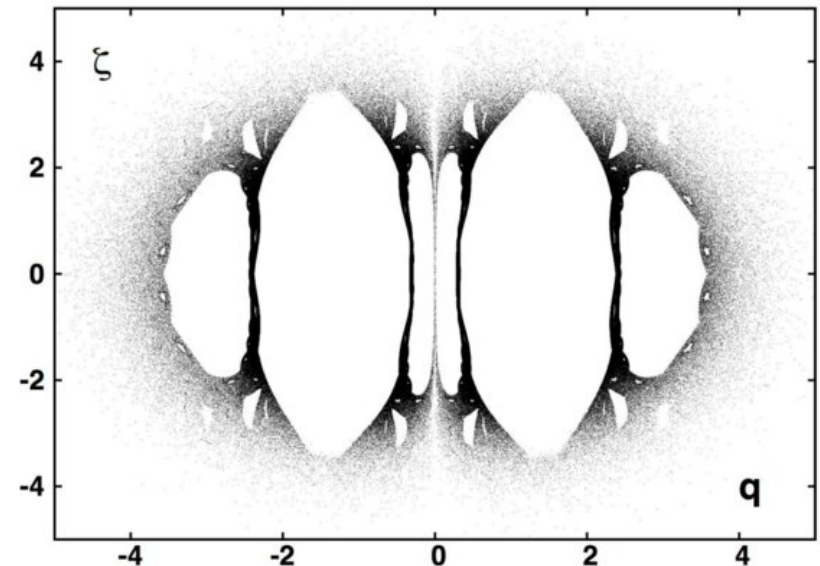
$$\dot{p}_i = \frac{\partial \mathcal{H}}{\partial q_i} = -q_i - \xi p_i$$

$$\dot{\xi} = p_i^2 - 1$$

$$m_i = k = T = 1$$

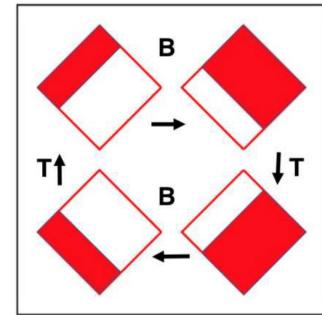
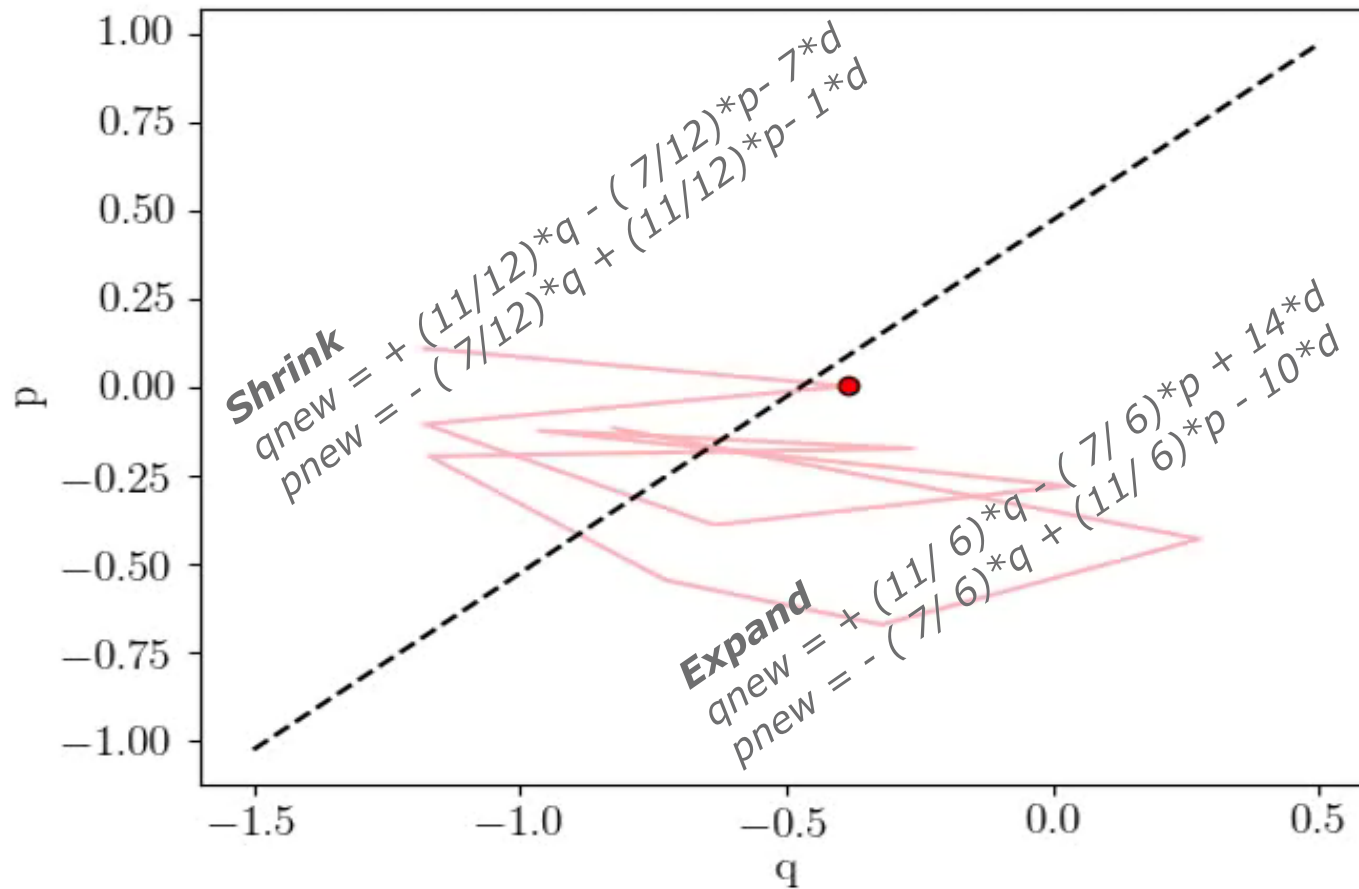
- The addition of a thermostat cause phase space volume to change

- Complex fractal patterns
- Unstable orbits in q - p - ξ phase space
- Linked to viscosity, entropy and maybe turbulence



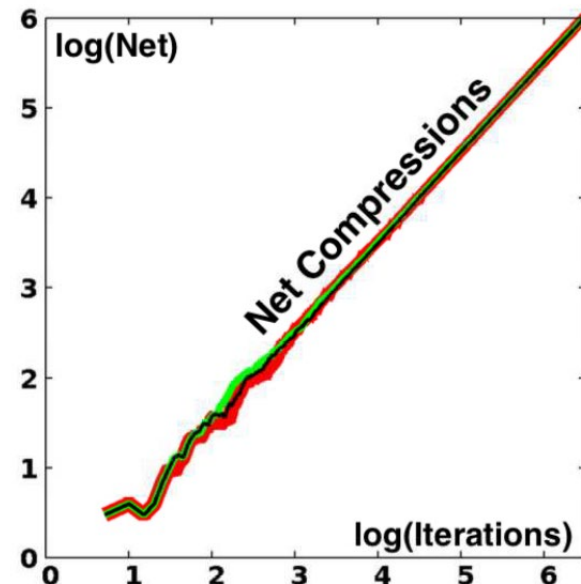
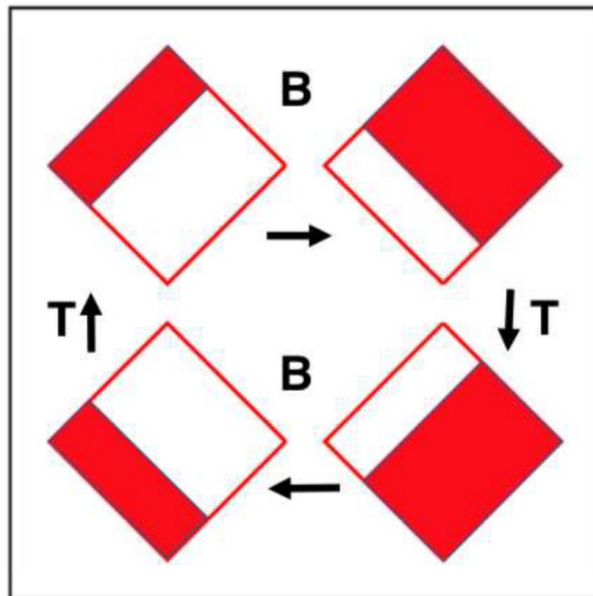
A Simpler System

- The baker map provides an easier way to explore the phase space dynamics of q-p system



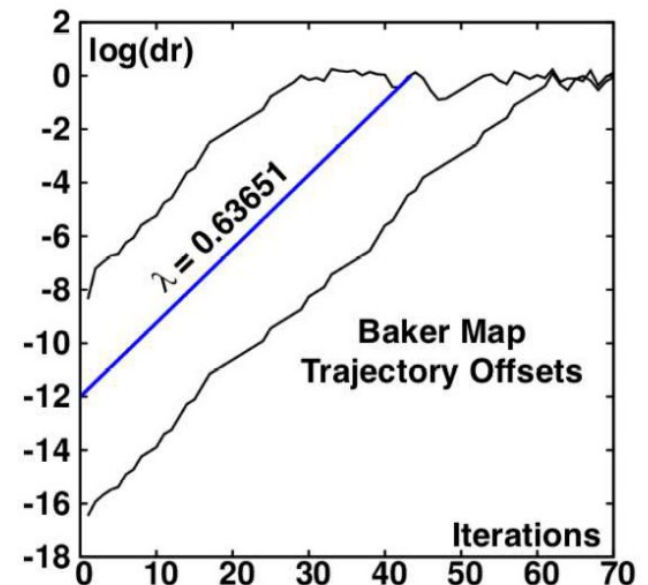
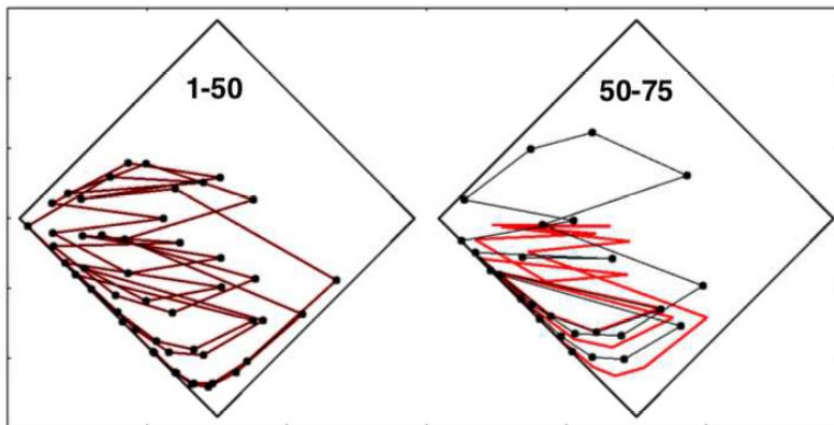
An Even Simpler System

- Shrinking phase volume and fractal phase-space
- On average a white area is halved two thirds of the time while a red area doubles one third of the time.
- The area decreases as $2^{(-t/3)}$ with t iterations
- Compressive iterations 1/3 of the total (varying precision)



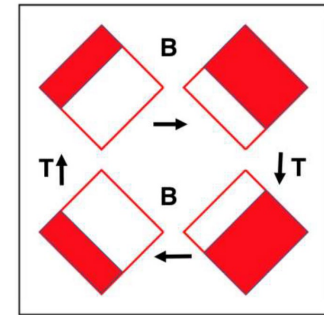
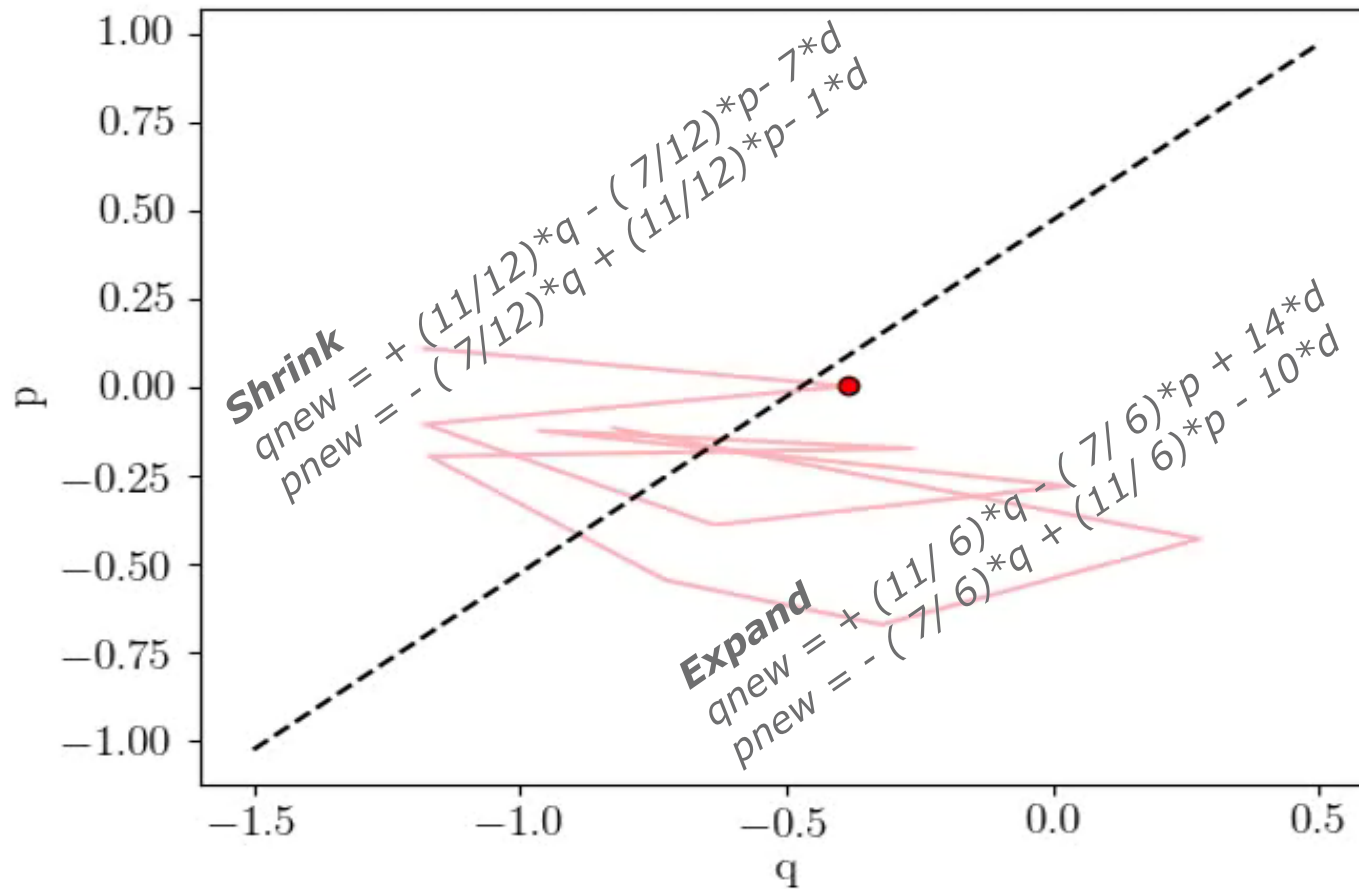
An Even Simpler System

- Chaotic behaviour seem in the trajectory divergence after about 40 iterations (comparing 64 vs 128 bit)
- Growth rates in the red and white regions, $\ln(3)$ and $\ln(3/2)$ give same result as largest Lyapunov exponent



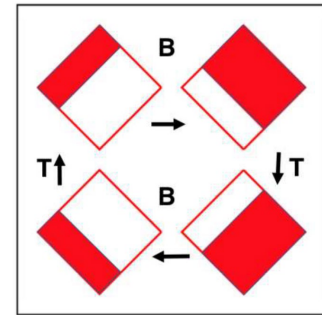
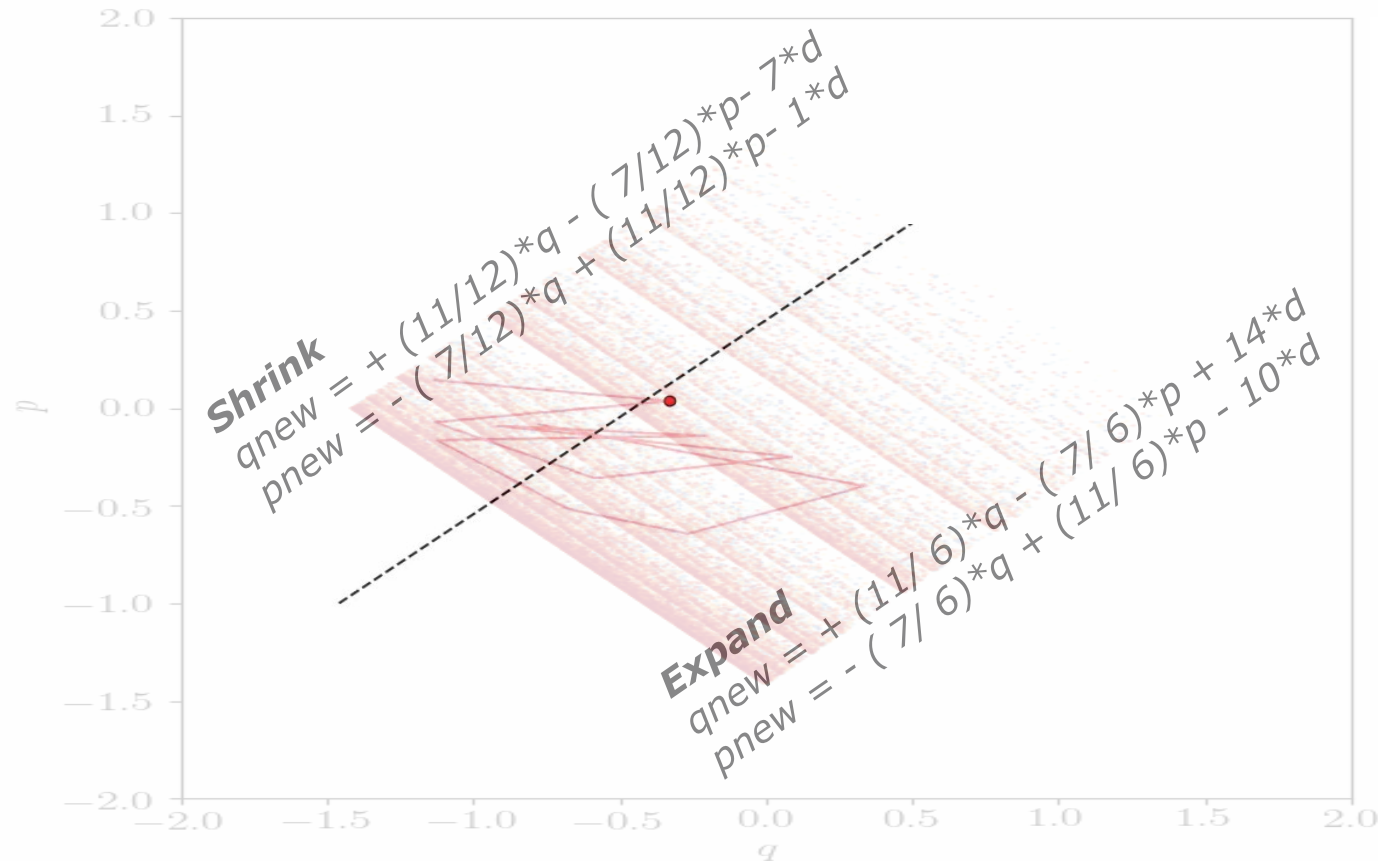
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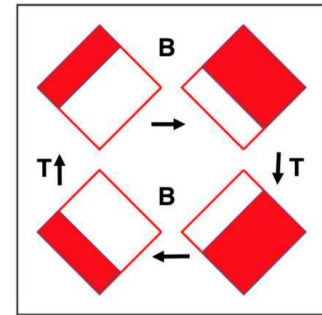
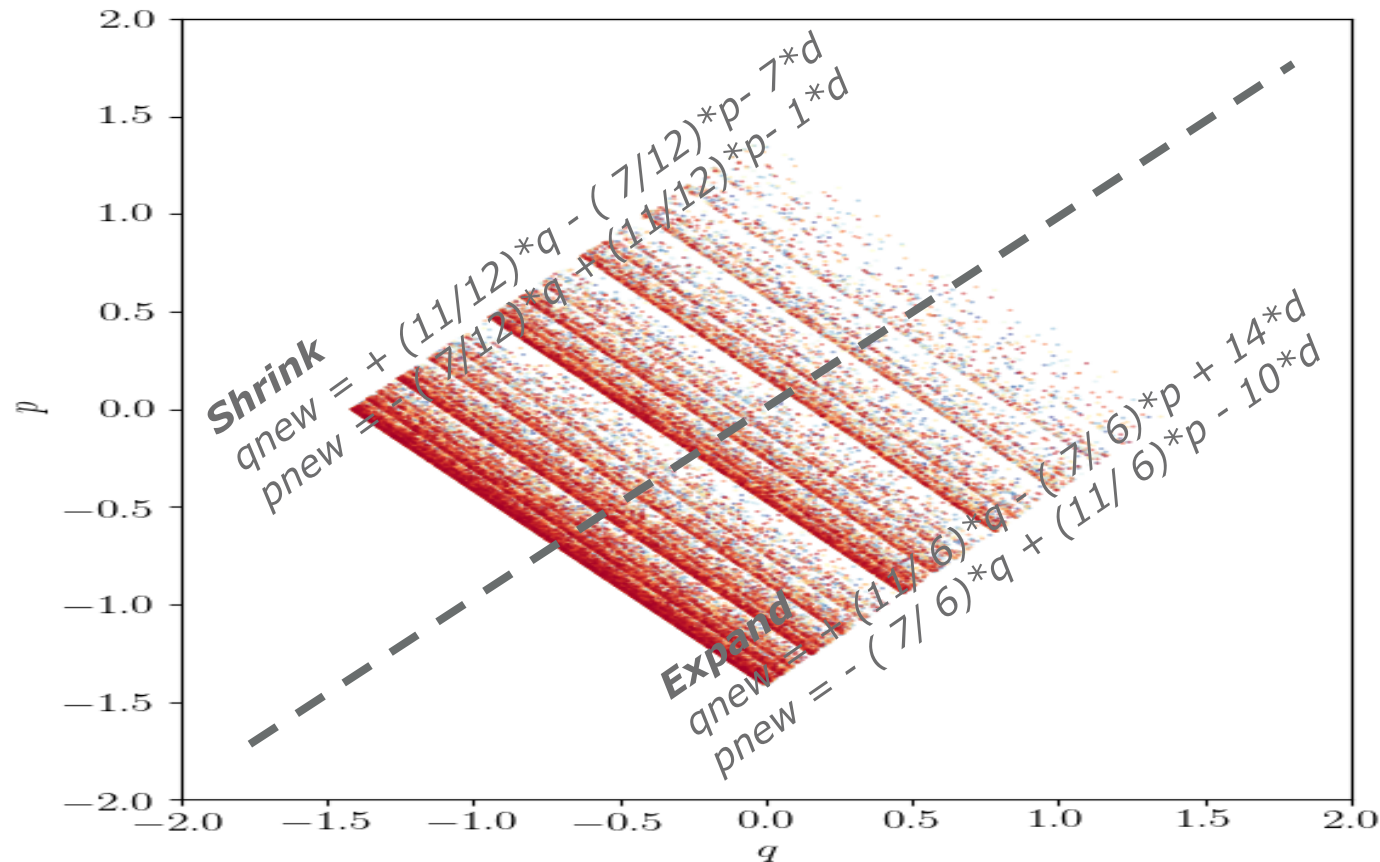
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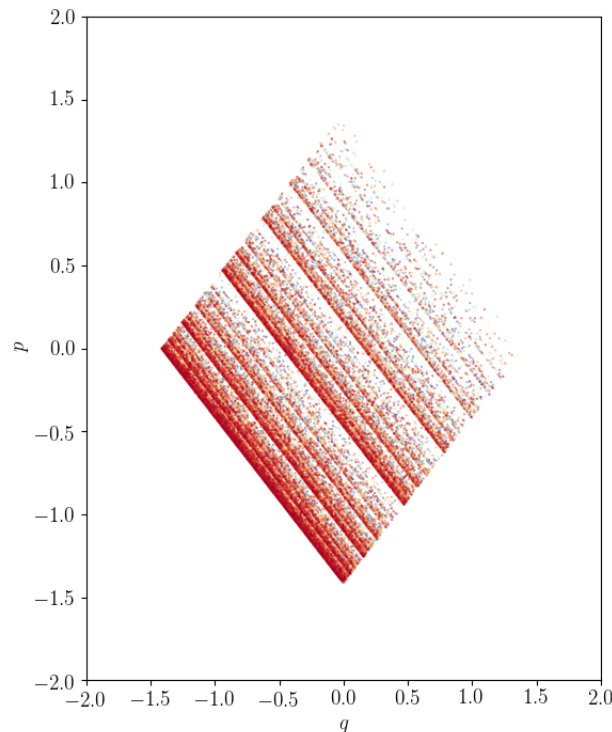
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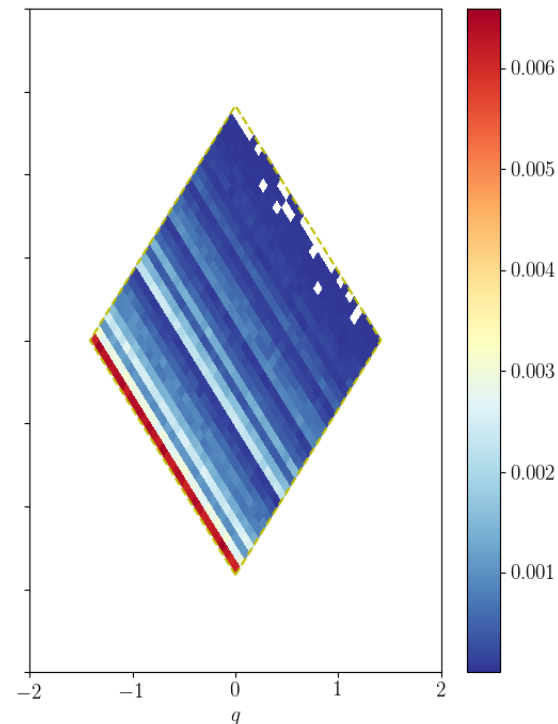
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Scatter plots over many iterations



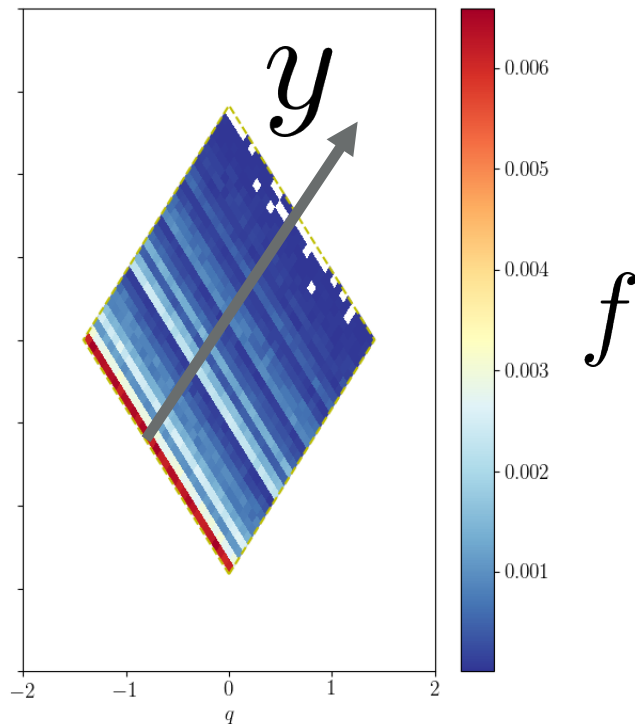
Binning probability of finding a q-p value (i.e. the phase space)



A Simpler System

- The binning is essentially 1D along the $q+p=\text{constant}$ line so we can map it to 1D

Binning probability of finding a q - p value (i.e. the phase space)

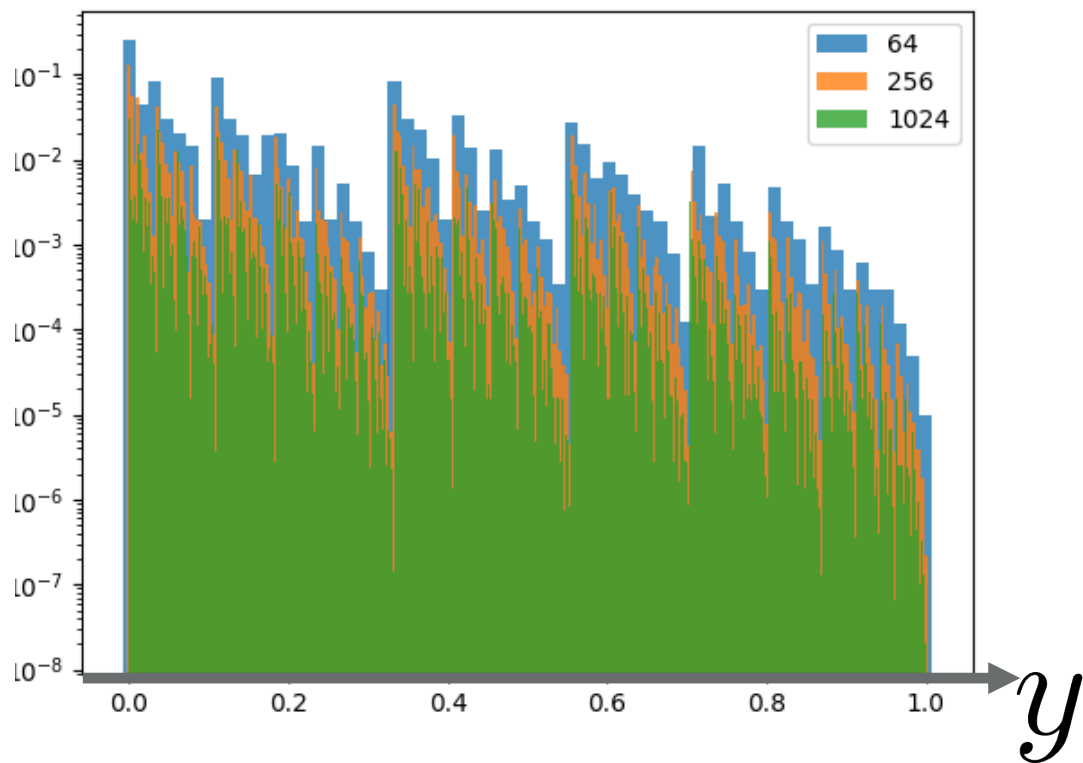
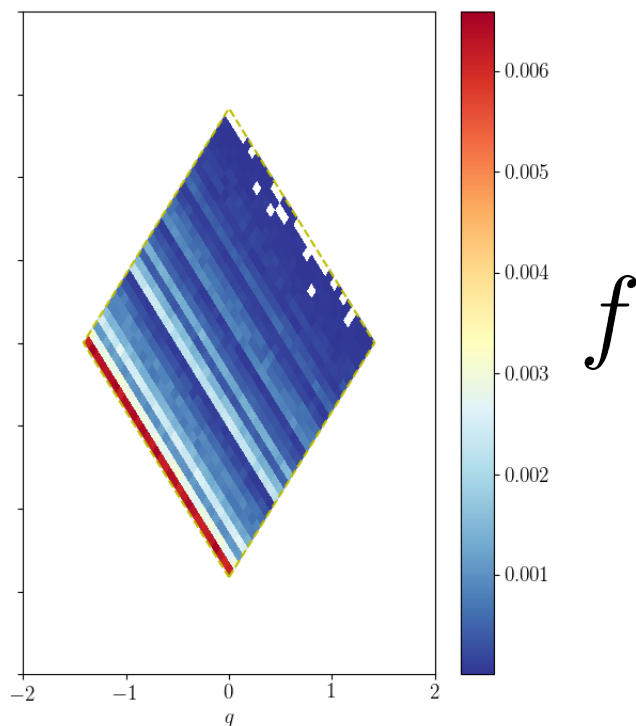


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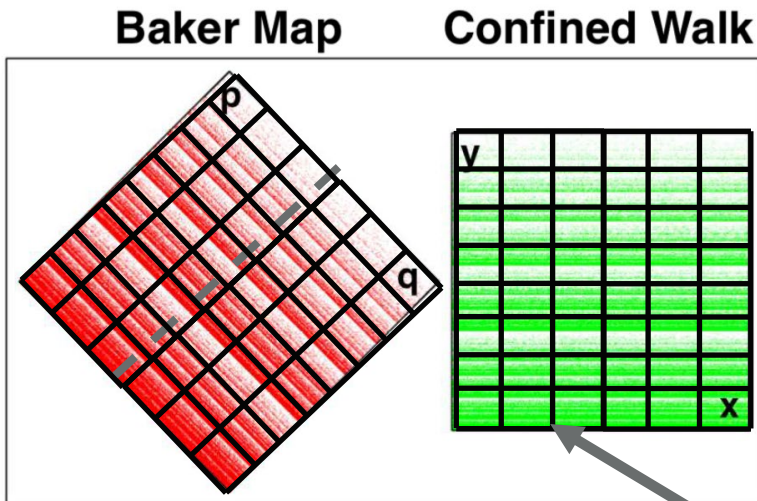
Binning probability of finding a q - p value (i.e. the phase space)

Mapped to 1D, log scale, varying bin resolution (64, 256, 1024 bins)



An Even Simpler System

- This 1D phase space system can also be generated directly from a random walk (y position and a random number)



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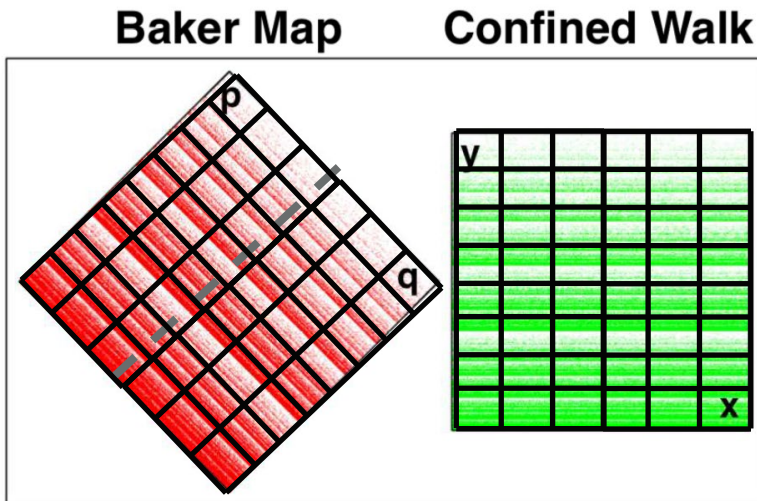
if (r < 1/3)
    ynew = (1+2y)/3
else
    ynew = y/3
end
    
```

Divide q-p positions (or y positions) into a grid and bin to get phase space.

We consider bins in multiples of 2 and 3

An Even Simpler System

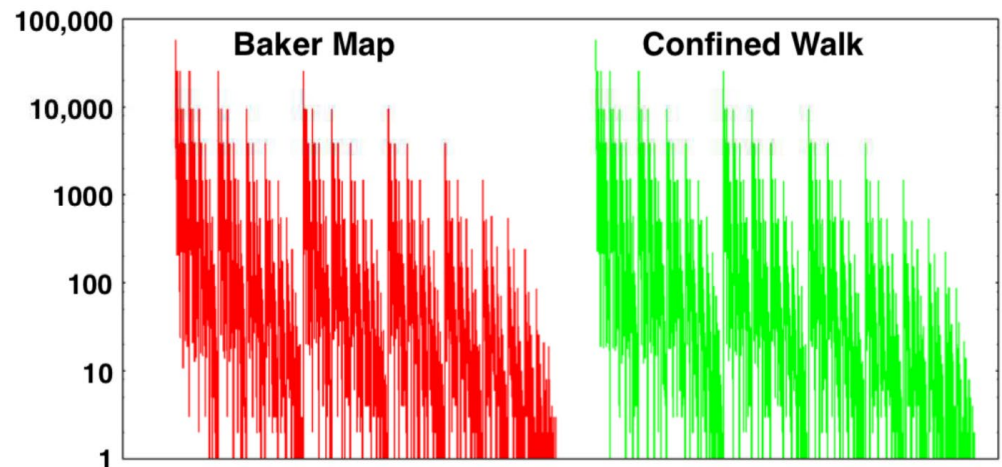
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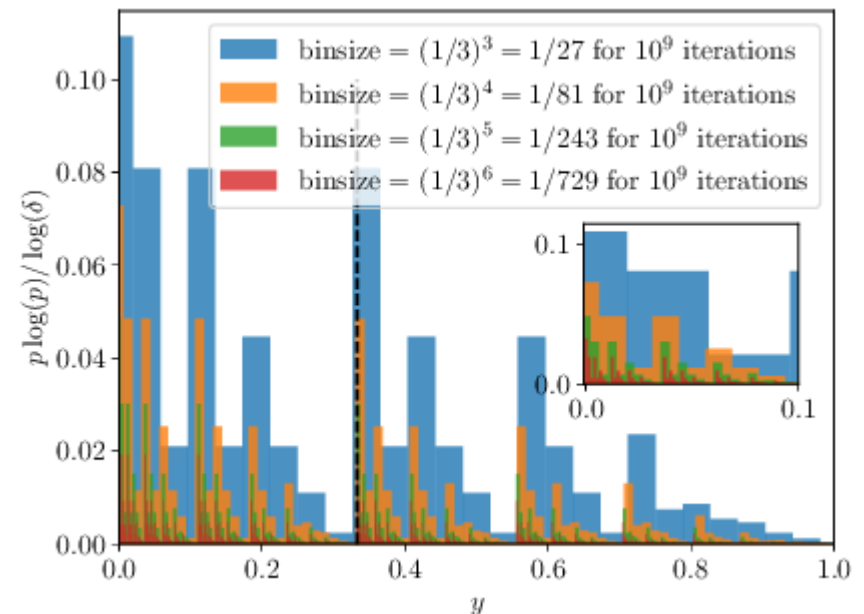
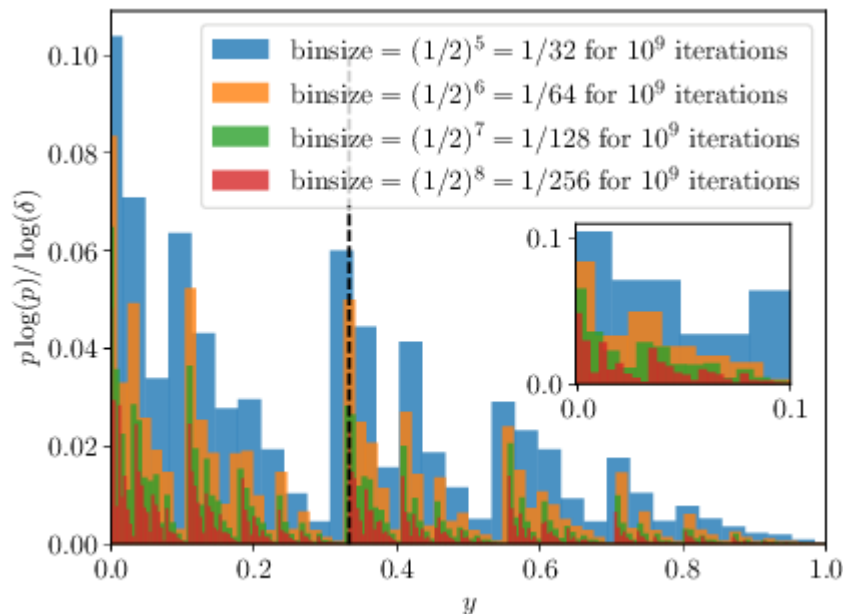
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- Identical phase space generated in both systems

Bin Resolution

- Depending on the binning resolution we get different phase space distributions from these systems
- Fractal and changes with binning resolution
- Using powers of 2 or 3 gives different results



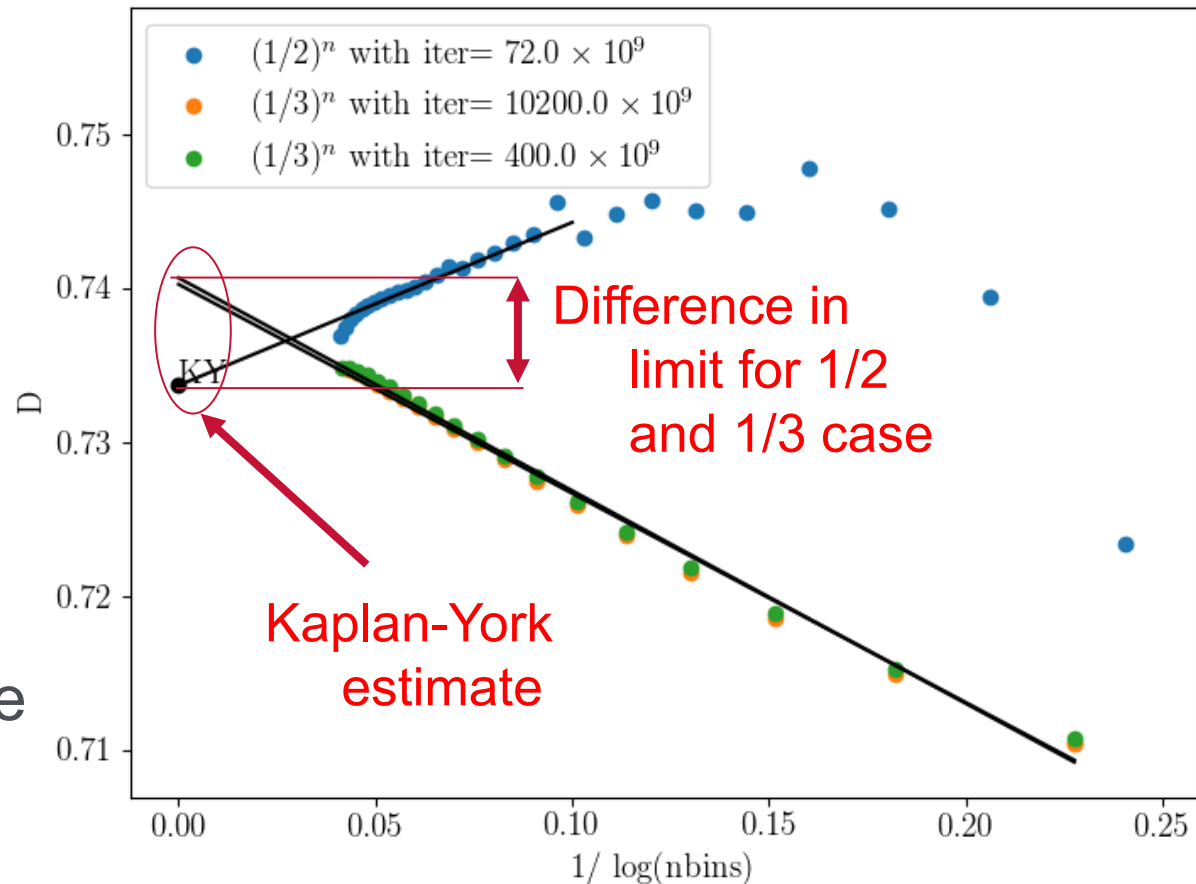
Bin Resolution

- Extrapolating to zero binsize gives fractal dimensions

- Fractal dimension estimated using the information dimension

$$D_I(\delta) = p \ln(p) / \ln(\delta)$$

- Here p is the probability (bar height) at a y value
- δ is the binsize



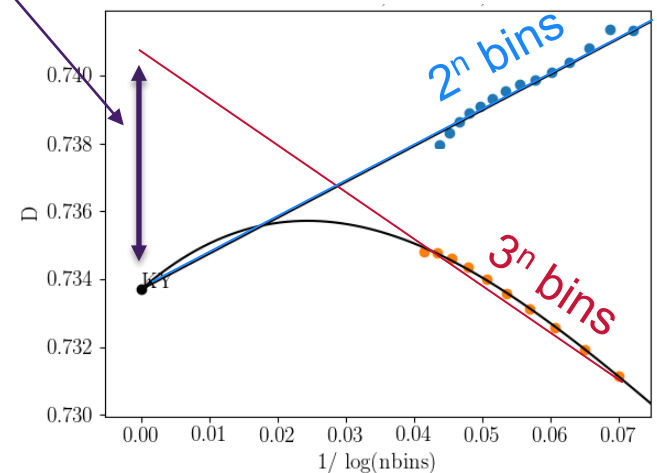
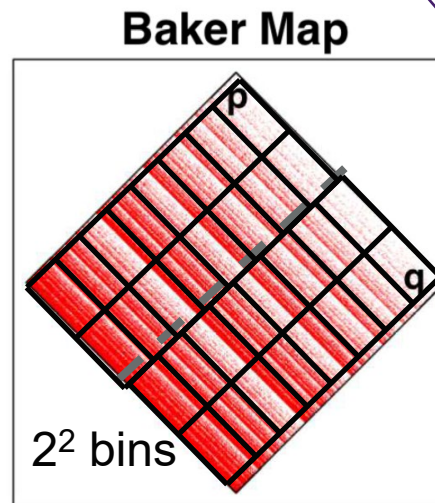
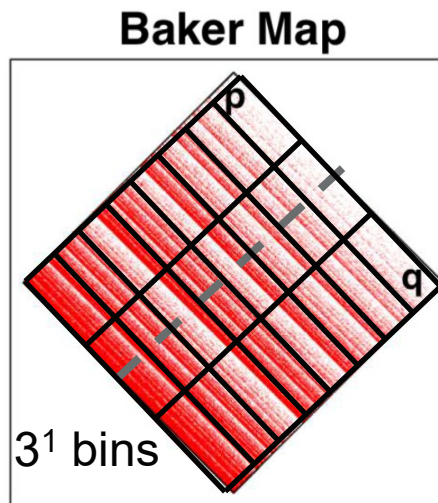
Summary

- Understanding phase space is essential to understanding NEMD systems and molecular fluid dynamics
- The Baker-map and random walk model phase spaces in a simpler way than continuous q-p NEMD, see:
https://github.com/edwardsmith999/fractal_phase_space
- The dynamics capture the loss of phase volume with interesting fractal distributions - linked to essential concepts like entropy, chaos and reversibility
- Different binning approaches result in different estimates for the information fractal dimension.
- Not clear why this is the case.
- More details: <https://export.arxiv.org/abs/2108.02251>

The Ian Snook Prize

\$1000 SNOOK PRIZES FOR 2021: The Information Dimensions of a Two-Dimensional Baker Map

For the simplest nonequilibrium Baker N2 Map these three approaches can give different results.



A pedagogical exploration and explanation of this situation is the 2021 Ian Snook Prize Problem.

Close

- We will aim to have another meeting next year – offers to host a meeting are always welcome (as well as funding)
- Please feedback suggestions and comments on this meeting and how we should develop the community

Thank you

Any final questions or comments